PLATFORM OVERVIEW

Balfour Beatty Signage Portal

A complete technical and functional overview of the digital signage procurement platform built for Balfour Beatty

APPLICATION OVERVIEW — MARCH 2026

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Platform at a Glance

01

A self-service web portal where Balfour Beatty site teams browse, customise, and order branded construction signage. Replaces spreadsheets, emails, and manual coordination with a single digital workflow.

125	27	324
PRODUCTS	CATEGORIES	SIZE & MATERIAL VARIANTS

Technology Stack

COMPONENT	TECHNOLOGY
Frontend	Next.js 16, React 19
Language	TypeScript 5
Styling	Tailwind CSS 4
Database	Supabase (PostgreSQL)

COMPONENT	TECHNOLOGY
PDF Generation	React PDF Renderer
Automation	Make.com Webhooks
Hosting	Vercel (Auto-deploy)
Image Pipeline	Python, OpenCV

> Fully branded for Balfour Beatty — navy and blue colour scheme, BB logos across all pages, emails, and generated documents. Every product image carries the Balfour Beatty emblem.

Architecture

The application is a monolithic Next.js app deployed to Vercel with server-side API routes connecting to Supabase PostgreSQL. Authentication is password-based with separate shop and admin tiers. External workflows are triggered via Make.com webhooks for email delivery and fulfilment system integration.

Browser
Next.js frontend
React 19

→

API Routes
Server-side
TypeScript

→

Supabase
PostgreSQL
Row-level security

→

Make.com
Webhooks
Email & routing

User Journey

02

From login to delivered signage — a streamlined digital workflow that replaces phone calls, spreadsheets, and email chains.

Login — Secure password → **Browse** — Categories & search → **Select** — Size, material & quantity → **Basket** — Review & adjust → **Checkout** — Contact, site & purchaser → **Confirm** — Order ref & email

Browsing & Selection

- 27 signage categories with visual product cards
- Full-text search across codes, names, and descriptions
- Product pages show all size and material variants
- Clear pricing per variant with image preview
- Products with custom fields capture site-specific data (site name, phone number, etc.)

Basket & Checkout

- Basket with images, quantities, and VAT breakdown
- Select or create a contact (name, email, phone)
- Select or create a delivery site (name, address)
- Assign a purchaser for PO approval
- Attach PO number and special instructions

Product Categories

Site Setup Pack
Environmental Signs
Site Entrance Signs
Information Signs
P.P.E. Signs
Hazard Signs
Fire Signs
Pedestrian Signs
Security Signs
Health Signs
Site Marking
Simple Safety Signs
Working at Height
Traffic Signs
Parking Signs
Roadworks Signs
Freestanding Signs
Display Boards
Notice Boards
Considerate Site Signs
Posters
Prestige Signs
Hoarding Signs
Finished Home Signs
Custom Signs
Extras & Accessories
Other Signs

> Materials include Correx, Foamex, Zintec, Dibond, and Self-Adhesive Vinyl — ranging from small 100×150mm labels to full hoarding panels.

Customisation Features

03

Beyond the standard catalogue, the portal supports fully bespoke signage and non-standard dimensions.

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Custom Sign Builder

Design bespoke signs from scratch. Choose sign type (warning, prohibition, mandatory, information, fire safety, directional, security, environmental), shape, material, and enter custom text. A live preview renders in real time. Submitted as a quote request.

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Custom Sizing

Enter non-standard dimensions on any product. The system automatically matches to the nearest standard variant for instant pricing, or flags for a manual quote if no close match exists. Area tolerance matching within 150%.

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Custom Fields

Products that require site-specific data (site name, phone number, address, etc.) present dynamic input fields at the point of ordering. Values are captured in the order and carried through to fulfilment.

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Pricing Logic

Standard items priced from catalogue. Custom signs submitted at £0 pending quote. Custom sizes auto-matched to nearest variant price. VAT (20%) calculated server-side. Prices validated server-side to prevent tampering.

Order Lifecycle

New — Order placed email sent → **Awaiting PO** — Purchaser approval → **In Progress** — Sent to production → **Completed** — Delivered & documented

Purchase Order Workflow

- Admin sends order to purchaser for PO approval
- Purchaser receives branded email with token-protected upload link
- PO document uploaded directly through the portal
- Order status automatically updated on upload
- Admin can download PO documents at any time

Document Generation

- **Quote PDF** — full order breakdown with branding
- **Order List PDF** — product codes with artwork status
- **Delivery Note PDF** — branded shipping document
- All PDFs generated server-side with React PDF
- Delivery notes can also be uploaded as files

Admin & Management

04

A dedicated dashboard gives the production team complete visibility and control over all orders across the entire estate.

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Order Dashboard

View all orders with status badges. Filter by status, contact, or site. Search by order number, contact name, or site. Expand any order to see full item details with product images and pricing.

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Status Management

Update order status directly from the dashboard. Statuses: New | Awaiting PO | In Progress | Completed | Cancelled

PDF

Document Actions

Generate and download Quote, Order List, and Delivery Note PDFs. Upload PO documents and delivery notes. All documents stored in the database and downloadable at any time.

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User Feedback

A suggestions widget lets shop users submit feedback and sign requests. Admin reviews submissions with a status workflow: New → Noted → Done / Dismissed. Counter badges show unread items.

Data Management

Reusable entities are managed via CRUD interfaces, eliminating repetitive data entry across orders.

Contacts

Name, email, phone. Upserted by email. Select or create inline during checkout.

Sites

Site name, full address. Upserted by name. Reused across multiple orders.

Purchasers

Name, email. Assigned to orders for PO approval. Receives automated email with upload link.

Database Schema

TABLE	PURPOSE	KEY FIELDS
bal_orders	Order headers	order_number, status, contact/site/purchaser, PO & DN docs, totals
bal_order_items	Line items	code, name, size, material, price, quantity, custom_data (JSONB)
	Reusable	

Integrations & Security

05

The portal connects to external systems for email delivery, fulfilment routing, and automated workflows.

Email & Automation

All transactional emails are triggered via **Make.com webhooks**, providing a flexible automation layer between the portal and downstream systems.

- **Order confirmation** — sent to contact on submission
- **Internal PO email** — full order details to production team
- **Purchaser notification** — PO approval request with upload link
- All emails are branded HTML with item tables, pricing, and images
- Custom sign items colour-coded by type in emails

Fulfilment Integration

"Send to Nest" action fires a webhook to route the order into the production system with complete order data, item details, and PO references.

Security

- Password-based authentication (shop & admin tiers)
- HTTP-only secure cookies with 30-day expiry
- Rate-limited login (5 attempts per minute per IP)
- Server-side price recalculation (prevents client tampering)
- Token-protected PO upload links (time-bound)
- Supabase Row Level Security on all tables
- Security headers: X-Frame-Options, HSTS, CSP

Hosting & Deployment

- Deployed on **Vercel** with automatic Git deployments
- Environment variables for all secrets and config
- PostgreSQL database hosted on **Supabase**
- Global CDN for static assets and product images
- Zero-downtime deployments on every push

> The Make.com integration layer means email templates, routing rules, and downstream system connections can be modified without code changes to the portal itself.

API Endpoints

Platform Summary

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The Balfour Beatty Signage Portal is a complete digital procurement system purpose-built for branded construction signage at scale.

Self-Service Ordering

Site teams browse, customise, and order signage without phone calls or spreadsheets

Brand Consistency

Every product image carries the Balfour Beatty emblem. All documents and emails are branded.

Custom Capability

Bespoke signs, non-standard sizes, and site-specific fields handled within the same workflow

PO Workflow

Automated purchaser approval flow with token-protected document upload

Full Visibility

Admin dashboard with filters, search, status tracking, and document management

PDF Documents

Server-generated quotes, order lists, and delivery notes with Balfour Beatty branding

Automated Workflows

Make.com integration for email delivery, PO routing, and fulfilment system connectivity

Scalable & Secure

Vercel hosting, Supabase database, server-side validation, and rate-limited authentication

> One platform. Visual product selection. Structured ordering. Purchase order workflow. Complete visibility from site to sign.

Get in Touch

For questions about the platform, feature requests, or a live demonstration, reach out to the team.

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